D 110236 **(Pages : 2)** **Name.....**

Reg. No.....

FIFTH SEMESTER (CBCSS–UG) DEGREE EXAMINATION, NOVEMBER 2024

Physics/Applied Physics

PHY 5B 08/APH 5B 08—OPTICS

(2019 Admission onwards)

Time : Two Hours Maximum : 60 Marks

The symbols used in this question paper have their usual meanings

Section A — Short Answer type.

Answer **all** *questions in two or three sentences, each correct answer carries a maximum of 2 marks.*

1. Define a thin lens.
2. Define first principal focus of a lens.
3. Define lateral magnification.
4. Explain interference.
5. Explain coherence.
6. Define Fiber Bragg Gratings.
7. Explain colours of thin film.
8. Write down the equation for resolving power of a grating.
9. Define Huygene's explanation of double refraction.
10. Define polarization.
11. What is hologram ?
12. Give two applications of holography .

(Ceiling 20)

Section B - Paragraph / Problem type

Answer **all** *questions in a paragraph of about half a page to one page, each correct answer carries a maximum of 5 marks.*

13. Explain the interference by a plane parallel film when illuminated by a plane wave.
14. Derive cosine law.
15. Write a note on non reflecting films.

Turn over

16. With necessary figure, explain interference by a plane parallel film illuminated by a point source.

17. Derive the equation for width of principal maxima of an N slit Fraunhofer diffraction.

18. Define double refraction. Explain Laurent's half-shade polarimeter.

19. With figure explain Fresnels half period zone.

(Ceiling 30)

Section C - Essay type

Essays - Answer in about two pages, any **one** *question.*
The question carries 10 marks.

20. Briefly derive and the intensity distribution of interference pattern

21. Explain with figure, single slit diffraction pattern. Explain the positions of maxima and minima

(1 × 10 = 10 marks)